Pixel Accuracy

You calculate pixel accuracy by dividing the number of correctly classified pixels by the total number of pixels in the image. This metric provides a quick global overview of your model performance.

Formula:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Practical Example:
You evaluate a medical scan containing 100,000 pixels. Your model correctly classifies 90,000 pixels as background and 5,000 pixels as a tumor. Your pixel accuracy is 95%.

Disadvantages:

- You get misleading results on imbalanced datasets.
- A model can predict 100% background on an image with a tiny object and still achieve 99.9% accuracy.
- You cannot evaluate the exact shape or boundaries of your predicted object.

Intersection over Union (IoU)

You use IoU to measure the spatial overlap between your predicted segmentation mask and the ground truth mask. You will often see this called the Jaccard Index. This metric penalizes false positives and false negatives heavily.

Formula:

$$IoU = \frac{TP}{TP + FP + FN}$$

Practical Example:
Your target object covers 1,000 pixels. Your model predicts a mask of 1,200 pixels. The exact intersecting area covers 800 pixels. Your IoU is 800 divided by 1,400, resulting in an IoU of 0.57.

Disadvantages:

- You penalize single-pixel errors much more heavily on small objects than on large objects.
- You struggle to optimize this metric directly using standard loss functions because it is not mathematically differentiable.

Dice Coefficient

You calculate the Dice Coefficient to measure the similarity between two spatial samples. This metric doubles the weight of the true positives. It behaves similarly to IoU but always yields a slightly higher

numerical value.

Formula:

$$Dice = \frac{2 \times TP}{(2 \times TP) + FP + FN}$$

Practical Example:
You use the same 800-pixel intersection from the previous example. The ground truth size is 1,000. Your prediction size is 1,200. Your Dice score is 1,600 divided by 2,200, yielding 0.72.

Disadvantages:

- You encounter the exact same small-object penalty problem seen in IoU.
- You can accidentally mislead stakeholders because Dice scores consistently look better than IoU scores for the exact same prediction.

Precision and Recall

You use precision to measure how many of your predicted positive pixels are actually positive. You use recall to measure how many of the actual positive pixels your model successfully found.

Formulas:

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

Practical Example:
You want to segment cars in satellite imagery. High precision means when your model flags a car, it is almost certainly a car. High recall means your model finds almost every car in the image, even if it falsely flags some empty roads as cars.

Disadvantages:

- You must constantly trade off one metric for the other.
- You cannot evaluate the overall quality of the segmentation mask using just one of these numbers independently.
- You fail to account for true negatives entirely.